

Livelihood Enhancement Through Improvement of Native Chicken Production and Food Value Chain Using SMART Technologies and Partnerships in the New Normal

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Rationale

The pandemic has placed an unprecedented burden on the world economy, healthcare, and globalization through travel, events cancellation, employment workforce, food chain, academia, and healthcare capacity. The COVID-19 pandemic has evidently disrupted the global food value chain with border closures, quarantines, and market, supply chain and trade interruptions. Nearly every aspect of our food system has been affected, causing bottlenecks that lead to food insecurity, inaccessibility and wastage. Improving the food supply and value chains in the new normal is imperative to adjust the management of the flow of goods and services towards increasing agricultural productivity and competitiveness.

Currently, the demand for native chicken meat is high because majority of the consumers prefer native chicken due to its distinct taste, unique flavour and texture and lower fat content over broiler chickens. According to PSA (2019) as of January 1, 2019, the total inventory of chicken was estimated at around 186.37 million birds where native/improved chicken contributed the highest share to the total inventory of 44.72 percent or equivalent to 83.34 million birds. From this inventory, Western Visayas nailed as the top 1 producing region with 12,883,030 or 15.46% of native chicken. The demand of native chicken in the region was influenced by the increasing number of restaurants serving native chicken specialty. Based on the records of Capiz Tourism (2020), there were 35 accredited restaurants and café in the province of Capiz and some of these restaurants serves the native chicken dishes like inasal, binakol, adobo, tinola and fried native chicken. These restaurants are commonly found along the main highway or national road to cater travellers. In Iloilo, the most popular among the native chicken restaurants are the *Tatoys Manokan* which serves the *best native chicken inasal* in the city, *Leah's Restaurant for Tinuom*, *Farm to Table Restaurant for Binakol* and *Y2K Restaurant* for native chicken adobo dish. Several restaurants are also found in other provinces of the region. However due to COVID 19, native chicken industry was greatly affected due to closures of this restaurants and the limited movement of people due to border restrictions.

In order to help native chicken industry, the DOST PCAARRD had launched a project on food value chain to enhance native chicken production in Region 6 by using the different SMART technologies that were developed already in some previous studies on Darag native chicken by different SUC's (CapSU, WVSU, CPU, ASU) in Region 6. Some of the technologies developed answer the gap in the production of native chicken which is high mortality due to diseases and low production rate. The following were the different research endeavors: To address the high chicken mortality, ethnobotanical anthelmintics were developed to control problems on gastro-intestinal parasitism (using Betel nut and Ipil ipil seeds powder) and the use of New Castle Disease vaccine to control the annual outbreaks of the disease. To lower the feed expenses, enhancement of the free range area by planting of some leguminous plants like mani mani (*Arachis pintoii*) and getting the ideal range measurement as to number of native chicken in the pen and the use of organic feeds. To increase the low hatchability rate of natural incubation of native chicken, artificial incubation was used using incubator and proper brooding

technique. The project also produced quality breeders, hardened chicks and slaughter native chickens to sustain the native chicken production in the region and to lessen disease outbreak, the use of Newcastle Disease vaccine. With these, the need for this project is to help the raisers to increase their production outputs using the technologies generated from the previous PCAARRD funded R&D projects thereby increasing the native chicken supply intended for use by the consumers and processors.

No available data was noted as to the processed products of native chicken that were sold in the supermarket and other stores except for the fresh vacuum packed dressed native chicken, live chicken in the wet markets during market day and native chicken specialty dishes in restaurants and kiosk. Hence, development of native chicken food products such as processed native chicken ready to eat specialty dishes, proper handling and preparation of vacuum packed choice cuts and whole dressed native chicken and other processed products in needed with the help of the Food Innovation Center under Food and Fisheries Research and Development Center (FFRDC).

Another project funded by PCAARRD in which the university handles is the CAPSU ATBI (CAPSULE). The TBI had conducted several capacity building training and marketing and financial management trainings as well. The TBI had conducted several capacity building training and marketing and financial management trainings as well. This TBI therefore is one of the partner of the project that helps incubatees in the successful native chicken production as source of their sustainable livelihood project.

On the other hand, Darag native chicken is one of the Philippine native chicken (*Gallus gallus domesticus*) which are originated in Western Visayas. This strain of native chicken is one of the major commodity in Region 6 which was given with more importance by DOST PCAARRD and had already established with the help of WVSU, CapSU, ASU and other SUC's. However, due to COVID 19, this industry was also affected.

Hence, this project will be one of the answer to help Darag native chicken raisers to improved their production as well as to add value through processing into food products with the use of SMART technologies and through partnership and linkages.

Objectives

Generally the project aims to conduct livelihood enhancement through improvement of native chicken production and food value chain using SMART technologies and partnerships in the new normal.

Specific Objectives:

1. To enhance native chicken production and food value chain using Smart technologies and partnerships as a sustainable livelihood and food resiliency in the new normal;
2. To increase and improve quality of native chicken production;
3. To produce fresh vacuum packed and ready to eat Darag native chicken and
4. Ensure a sustainable food value chain for native chicken in the region.

Project Description

The Covid-19 pandemic has forced everyone to employ the "new normal". On May 27, 2020, the Philippine House of Representatives approved House Bill 6623 or the "Better Normal for the Workplace, Communities, and Public Spaces Act of 2020," which standardized the basic safety measures aimed at reducing the risk of virus transmission within workplaces and public spaces. In compliance with these regulations, the project team ensures that all of its activities are well-planned and aligned with the requirements of the "new normal" set up. These was documented as basis for modifying future project implementation, as well as developing policy recommendations and sustainable food value chain operations.

A. Pre – Implementation Activities. Prior to the full blown project execution, pre-implementation activities, such as signing of the Memorandum of Agreement (MOA), organization of the project team, hiring of project staff, issuance of special orders to participating stakeholders, and conduct of inception meeting was carried out. All stakeholders who involved in the value chain process were identified and organized. Communication to the DA Regional office 6, provincial office down to its municipal office were done to inform them about the project and a possible partnership. An orientation about the project were done to brief the participating entities and encourage their commitment to join the undertaking.

B. Implementation Activities

- 1. Reconnaissance Study.** The project team conducted a reconnaissance study or the preliminary evaluation of all existing information, including the actual production and processing practices of the cooperating farmers and food processors and the prevailing activities of key stakeholders. Results of this undertaking served as basis for identifying and prioritizing the best interventions that were applied for each of the participating stakeholder. Also, the particular activity helped establish the baseline data where the project's performance were measured on. The study were done within two to three weeks from the official start of project implementation.
- 2. Community Organizing.** Community organizing and enterprise building are two major components of the project. Specifically, these activities forms part of the establishment or enhancement of the S&T community-based farms (STCBF) and agri-based technology enterprises hosted under the ATBI program.

The project were implemented in selected sites within the region. STCBF project sites were done in Dumarao, Mambusao and Dumalag, Dao, Cuartero, Tapaz, Sigma and in Roxas City, Capiz. The project team identified farmer co-operators with the help of the Municipal Agriculture office of the selected towns of the project. The priority co-operators of this project are those farmers who are beneficiaries of native chicken dispersal of any government agencies and or NGO's (ex. YRRP of Department of Agriculture), retired employee and pandemic affected business personells and OFW. Partnership with the private native chicken group in the island (Panay Darag Breeders Association) was done. Those raisers who are chosen to be a STCBF co-operator were trained for the package of technologies that improves their native chicken production to ensure the sustainable supply of native chicken within the region. (Proper health protocol and precautions by wearing facemask, social distancing and provision of hand sanitizer or alcohol were strictly employed during the meeting/ training). All available technologies generated from the previous PCAARRD funded R&D projects on native chicken were utilized. The selection of appropriate technologies were identified by the project team in consultation with PCAARRD and technical experts. The initial lists of technologies for this project are listed in Annex 1. These technologies include both Intellectual property-based and extension related technologies. These technologies were utilized for extension to native chicken raisers that were involved in the production aspect of the project. The project team also identified incubatees and trained for the production, product processing and marketing of native chicken. For technology support, PCAARRD, the DOST through CAPSU ATBI provides the needed assistance to the incubatees.

These STCBF raiser co-operators as well as the incubatees for processing were organized to form an association which were applied for registration in DOLE.

3. **Capability Building.** Capacity building activities were also be offered through the ATBI for business development related topics, while technical experts were tapped to facilitate trainings related to the application of technologies at the farm enterprise level. Technical trainings on the application of smart technological interventions (i.e., production management on housing and range, dewormer, vaccine, incubator, etc.) is a major undertaking of the project.
4. **Production Activities.** Native chicken production in the Philippines is categorized as backyard poultry production. One of the major constraints in native chicken production of most backyard raisers is the low production outputs due to high mortality caused by certain diseases, high cost of commercial feeds and other veterinary drugs. In order to answer this problem and to improve native chicken production of the raisers, introduction of the technologies for native chicken production developed for Native Chicken of the PCAARRD were introduced to identified raisers/farmer cooperator through trainings and seminar. Some of these technologies are: proper housing and range area management, feeding management (using several low cost feeds), health management (vaccination, deworming and with the use of pro biotics/ supplements), breeding (selection, Artificial incubator), brooding and hardening management. It is a package of technologies that were developed by host agency (CapSU), WVSU, ASU and DA Cebu and CPU and funded by PCAARRD. To further increase the production, all beneficiaries were organized and were formed to one Capiz Darag Growers Association(CADAGA) which is now a DOLE registered native chicken association in the province of Capiz.
5. **Processing.** As to date, no processors for native chicken were recorded in the region. Hence the project focused on proper handling and packaging of fresh vacuum packed choice cuts and whole dressed native chicken meat that were sold in restaurants, hotels and some households.
6. **Marketing.** Marketing is also a key component for achieving a sustainable food value chain. A full package of training to ATBI incubatees were done from production up to marketing of their products. Technical experts for marketing were tapped by the ATBI to guide the incubatees in marketing their products. The key customers of the native chicken products are the restaurant's and hotels serving native chicken menu. Assistance from the different government agencies to help in the marketing of the native chicken products were also prioritized in this project (DTI, BIR, LGU). Online marketing and promotion through social media were also be employed.
7. **Partnerships and Linkages.** Supply chain management puts emphasis on collaborative relationships among stakeholders. Strategic partnerships and institutional linkages are important factors for integrating the overall operations and sustainability of the food supply chain. Hence, a major undertaking of this project is to enhance partnerships and linkages among key players of the food system. Potential partners who can provide long term financial, technical and business development-related support were tapped to support the operations of the project co-operators and the ATBI.

The project forge new tie-ups with farmer's associations (like PADABA – Panay Darag Darag Breeder Association and Capiz Darag Growers Association), the Department of Trade and Industry (DTI) Regional offices, Department of Agriculture, non-government organizations (NGOs), business organizations/chambers, state universities and colleges (SUCs), and research and development institutes (RDIs).

The project also tie up with the CHED funded research (RISE Capiz) in helping Riverine Communities for Darag native chicken production.

8. **IEC Development and Technology Promotion.** IEC materials were developed to increase awareness and promote these initiatives to the public. This were filed for copyright protection at the Intellectual Property Office of the Philippines.
9. **Policy recommendations / Policy Inputs.** Policy recommendations and inputs were prepared to address underperformances, establish a more dynamic supply chain processes and form the strategic directions that will govern the plans for sustaining the project particularly on the vaccination schedule and the standard pricing of native chicken.

Monitoring and Evaluation. Regular monitoring and evaluation of the food value chain project, including its components and participating stakeholders were done to ensure that all activities and accomplishments are on track vis-à-vis the set goals and objectives. The food value chain performance were measured and analysed. The results of this analyses will serve as basis for modifying and improving the overall implementation of the system. Qualitative and quantitative assessments of effects and outcomes of S&T interventions were also done. Field visits, regular face-to-face meetings and preparation of periodical written reports are among the modes for monitoring the project. Virtual meetings and other digital based M&E systems were also considered while travel restrictions are in place due to Covid 19 pandemic.

Sustainability plan. Sustainability of the food supply chain is mainly driven by market demand. Hence, the project has to ensure that marketing related activities are constantly done by the project. A strategic management plan has been prepared as well to guide the project team and all participating stakeholders in long term implementation of the system. Institutionalization and commitment of the host agency and the participating stakeholders to continuously pursue these initiatives are also important aspects to ensure sustainability. In addition, the TBI was continuously collect reasonable amount of fees from incubatees for services rendered. The project cannot rely solely from funding agencies to sustain the operations of the TBI, hence partnerships and institutional linkages have to be maintained to support counterpart funding.

Target Beneficiaries

- Community-based farm enterprises
- Displaced OFW
- Business owners affected by Covid 19
- Technology-based startups/spinoffs
- Farmers and fisherfolk, cooperatives and associations
- Industry (Traders, Retailers)
- General public/consumers
- Local government units
- National government agencies
- Non-governmental organizations

Results of Interventions

The table below shows the different results and actual accomplishments of these project.

Expected Output	Details/ Activities Conducted	Target	Actual Accomplishments		Remarks
			No.	%	
People Services	Organize and provide support to the farmer cooperators.	100	224	224%	Organized Capiz Darag Growers Association (CADAGA) and Ivisan Darag Native Chicken Growers Association Provide support to IFALGAM
	Train the farmer cooperators on business management, package of technologies and Smart technologies (production of native chicken)	100	334	334%	
	Train food processors on processing of native chicken in vacuum packed and choice cuts	10	10	100%	
	Support 2 new incubatees	2	4	200%	The incubates are enrolled and mentored under ATBI
Products	Produce (20,000) heads of slaughter native chickens	20,000	21,800	109%	This data were all taken from different farmer cooperators and the STCBF
	Process (5,000) kgs of native chicken meat fresh vacuum packed choice cuts/ whole dressed and canned/bottled ready to eat specialty native chicken dishes	5000	6,537.28	130.75 %	Majority of this production were coming from supported

					STCBF farms and farmer cooperators and CADAGA. This processed DNC were sold to restaurant and some hotels such as El Circulo Hotel, Gabriels, Tumandok and some resto in Roxas City. The project also had developed some ready to eat food products such as Hamonado, adobo, binakol. Training on Hala Meat production was also done.
	Use 5 POTS and SMART Technologies at the farm level and 1 in processing enterprise	6	6	100%	
Places and Partnerships	Forge 10 linkages and partnerships to LGU's, farmer associations, DOST Regional offices, DTI Regional offices, NGO's, business organizations/ chambers, SUCs and RDIs and MOA with PCAARRD/ DOST	10	15	150%	PADABA, CADAGA, DA RFO 6, DOST 6, 13 municipalities, IFALGAM, WVSU, DA RFO 7, Deala Integrated Farm, Nong Noys Farm, DVTS, etc
Publications	Develop 1 promotional videos for the STCBF enterprises	1	2	200%	

	Develop/ produce 4 Information, Education and Communication (IEC) and promotional materials for STCBF enterprises and processor	4	6	150%	
	Update 1 TBI Operation Manual	1	1	100%	
	Update 1 TBI Curricula	1	1	100%	
	Develop 1 Publication on gender responsiveness of the project	1	1	100%	
	Prepare/ update 2 incubatee business plan	2	4	200%	
	Develop 2 Publications (English and laymanized) on Best practices and outputs of the farm clusters involved in Darag food value chain project and sustainable business models for community based enterprises	2	4	200%	
	Develop 1 Food value chain sustainability plan	1	1	100%	
Patents/IP	File 5 copyrights for the developed IEC materials and videos.	5	6	120%	
Policy	Develop 1 policy input in support of the food value chain operations	1	2	200%	
Impact to Community	<i>Social Impact:</i> • More available and accessible fresh and processed food products • Technology transfer and commercialization strengthened • Strengthened Public-Private Partnerships • Innovation ecosystem improved • Technopreneurship empowered • Local agribusinesses supported	The project had helped some restos in the province to have access in fresh vacuum packed Darag native chicken particularly on El Circulo Hotel (Cocina and Zugbao), Tumandok Restaurant, Gabriels and some households. The project also participated in Capiztahan Manindahan event. This project also help some ex OFW who were affected by the pandemic to have livelihood income and some retired government and pandemic affected business owners. Showcasing of the food products of Darag native chicken from this project was also done			
	<i>Economic Impact:</i> • Income generated by farm food producers / STCBFs	The project helped some of the raisers to increase their production by decreasing mortalities			

	30% increase in the volume of production through the enhancement of the food value chain	caused by diseases using technologies like vaccination, deworming and the use of some botanicals to prevent disease. The project also increase their income by lessening their cost of feeds through using of the naturally occurring feedstuffs in their areas.	

Lessons Learned

This project given by DOST PCAARRD is a great venue that helps a lot native chicken industry in the region despite of COVID 19 pandemic. Through this project, we have learned that theres a very big demand on native chicken industry particularly from different restaurants that serves native chicken as their specialty. The demand for fresh vacuum packed native chicken in one of the best hotel in Roxas City is 100 heads per week which shows that native chicken farming has a potential to be one of a sustainable livelihood source especially during pandemic. Many people are very interested on native chicken farming especially those pandemic affected individual (business owners, OFW and some retired employee). The demand of native chicken is very high not only in restos but also for the Department of Agriculture for their dispersal program as source of livelihood of those affected by typhoons, floods and even today the ASF outbreak in hog raisers. In every gathering and occasions, native chicken menu is always the center of the food preparation. Native chicken are also very prone to diseases, hence prevention of using technologies developed is strongly needed. In addition, feeds is the main problem of many native chicken raisers. However, this can be minimize with the help of using some alternative feedstuffs which can be found in their localities such as tricanthera, azolla, mani mani, banana trunk, camote and kangkong.

Recommendations

The project implementors would like to recommend that this extension activity will be continued and sustained since the demand for native chicken is very high. All projects cooperators as well as the organized farmer association(CADAGA) will be follow up regularly. Close monitoring and updates should be done to all members especially the ATBI incubatees (Gallaza Integrated Farm, Esher Tiva Integrated Farm, MJ Darag Native Chicken Farm and Contreras Darag Native Chicken Farm), the established STCBF farms and all farmer cooperators. Proper training should be done to ensure that all raisers will apply the proper production protocol on native chicken farming. Development of ready to eat Darag native chicken product should be conducted to have access of this DNC food products particularly in some Pasalubong Centers. Registration to Halal Meat through DA Region 6 will be continued. More collaboration and linkages to other agencies and institutions will also be conducted.

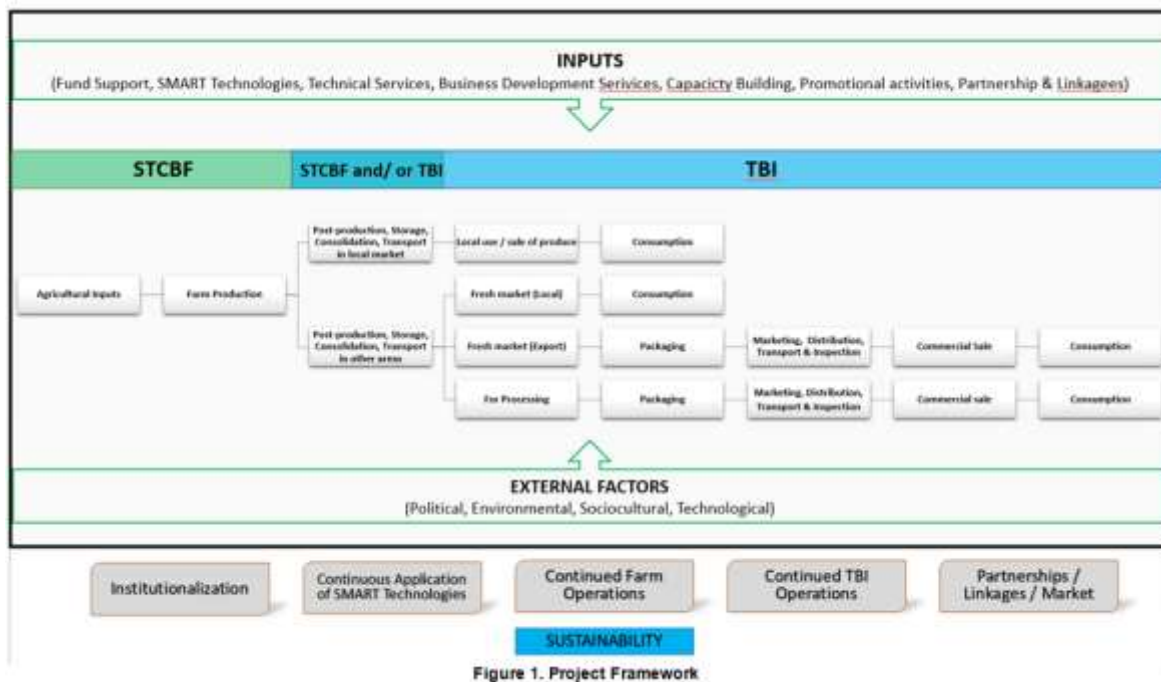


Figure 1. Project Framework

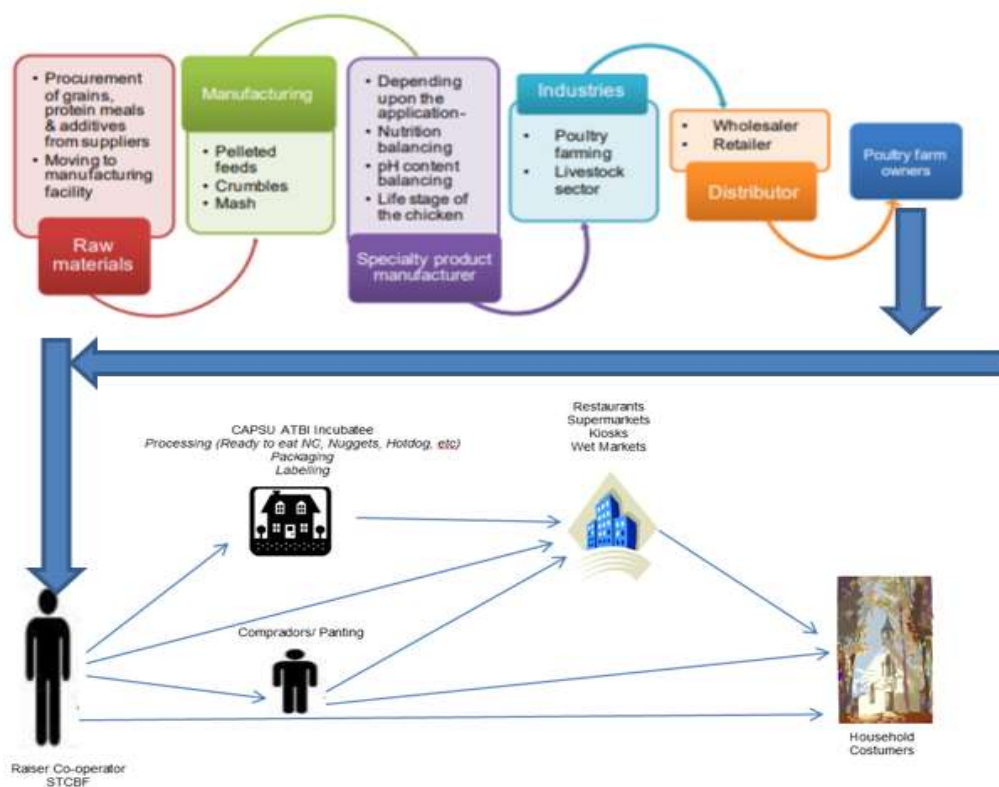


Figure 2. Supply Chain Framework of Native Chicken

Annex 1. List of technologies on native chicken

✓ Production of Native Chicken (SMART Technologies)

- Ethnobotanical Anthelmintic (Gastro intestinal Parasites)
- Oil Emulsion New Castle Disease Vaccine
- Artificial Incubator
- Range Enhancement/ Organic Feeds
- Breeder Selection
- Brooding and Hardening Management

✓ Processing

- Processing, Packaging and Labelling of choice cuts and whole dressed native chicken (vacuum pack)
- Processing, Packaging and Labelling of ready to eat native chicken specialty menus

Organized Capiz Darag Growers Association (CADAGA) and Ivisan Darag Native Chicken Growers Association Provide support to IFALGAM

